

Education

- 2023-present **University of Waterloo**, Combinatorics and Optimization, Waterloo, ON, CANADA
PhD, Combinatorics and Optimization. GPA 88.0/100.0
- 2021-2023 **New York University**, Courant Institute of Mathematical Sciences, New York, NY, USA
Master, Mathematics. GPA 3.8/4.0
- 2017-2021 **University of Michigan Ann Arbor**, College of Literature, Science, and the Arts, Ann Arbor, MI, USA
Bachelor of Science, Honor Mathematics, Computer Science. GPA 3.7/4.0

Research Experience

- 2023-present **Critical Points at Infinity, University of Waterloo**. Supervised by Stephen Melczer
 Using toric variety to explore proper compactification reflecting the properties of infinity points of algebraic varieties.
- 2023 **Distance Measurement of Mesh Surfaces, New York University**. Supervised by Daniele Panozzo
 Proposed and implemented algorithms determining whether a surface deviates a certain distance from another surface in 3-dimensional Euclidean space.
- 2022 Summer **Algebraic Geometry Solver, Google, Inc.** Supervised by Todor Milev, Michal Cierniak
 Worked in Learning & Education, Learning Platform, Math team; Implemented polynomial divisions in computer algebraic system supporting the math solver, which will be launched in Google Search; Realized end to end algebraic geometry solver using Wu's Method, serving as a complementary method of rule-based solver for automatic geometry theorem proving.
- 2019-2020 **Origami on Lattices, University of Michigan Ann Arbor** Supervised by Ian Tobasco, Harry Richman
 Quantitatively characterized the moduli space of configurations of an isolated origami hexagon. Formulated energy bounds for hexagon configurations which are "near unfolded". Described a combinatorial structure on the space of deformations of a regular origami hexagon, under the action of flip operators. Preprint, *Origami Hexagon Deformations and Flip Graph*.
- 2020 Summer **Efficient Eigensolvers and Applications, ICERM, Brown University** Supervised by Yanlai Chen
 Focused on numerical methods on eigenvalue problems and their applications. Implemented page rank algorithms and applied on website domains to obtain webpage rank scores; Developed performance comparison code frames; Built up web crawlers with directed graph generators.
- 2019 Summer **Natural Language Processing Optimizer, Facebook, Inc.**
 Worked in Facebook Artificial Intelligence Applied Research Organization, Language and Translation Technologies team; Maintained and optimized PyText, an NLP modeling framework based on PyTorch; Implemented mixed precision training; Designed and optimized memory-efficient fp16 optimizers, and established benchmark for fp16 training on models as NewBert, XLM.

Publications & Presentations

- CAAC *Critical Points at Infinity in Analytic Combinatorics*, Combinatorial Algebra Meets Algebraic Combinatorics, Halifax, January 2026. Talk.
- JIS *An exceptional equinumerosity of lattice paths and Young tableaux*, co-authored with Liam Ayres, Evan Bialo, Aidan Cook, Alwin Chen, Mattheus Froese, Maryam Mohammadi Yekta, Oliver Pechenik, Benjamin Wong, Journal of Integer Sequences, Vol 28, 2025.
- OGMC *Critical Points at Infinity in Analytic Combinatorics*, Ontario Graduate Mathematics Conference, Totonto, June 2025. Talk.
- Master Thesis *Optimizing Surface Closeness Measurements: An Analysis of Distance Field and Adaptive Simplification Methods in Tetrahedral Meshing*, 2023.
- Undergraduate Thesis *Deformations of the Origami Hexagon, Flipping Operations, and Pythagorean 4-Tuples*, co-authored with Harry Richman, 2021, preprint.
- SIAM *Origami Hexagon Deformations and Flip Graph*, SIAM Annual Meeting, 2021.

- YMC** *Origami Hexagon Deformations and Flip Graph*, Young Mathematicians Conference, Ohio State University, virtual, August 2020, Talk.
- UMS** *Origami On Lattices*, Undergraduate Mathematics Symposium, University of Illinois Chicago, Chicago, IL, November 2019, Poster.
- SPIE** *U-Net-Based Deep Learning For Detection And Segmentation Of Vertebral Bodies In Spinal MRI Scans*. SPIE Medical Imaging, Huston, TA, 2020.

Affiliations

- 2024-present **Women in Mathematics**, University of Waterloo
- 2021-2023 **Association for Women in Mathematics (AWM)**, NYU Courant
- 2020 **Women in Science and Engineering**, University of Michigan
- 2018 **University of Michigan Programming Team**
Participated in the 2018 ICPC East Central North America Regional Contest.

Honors & Fellowships

- 2026 Graduate Student Research Dissemination Award
- 2024, 2025 Women in Mathematics Mentorship Award
- 2022 Summer@ICERM Reunion Funding
- 2021 SIAM Annual Meeting Student Travel Award
- 2020 Summer Undergraduate Research Fellowship
- 2017-2020 University Honors

Teaching/Mentoring Experience

- 2024 Spring **Directed Reading Program: Computational Algebraic Geometry**, mentor, University of Waterloo
- 2023 Fall **Introduction to Combinatorics**, tutorial leader, University of Waterloo
- 2022 Fall **Honor ODEs**, recitation leader, New York University
- 2021 Summer **Number Theory**, math counselor, Ross Math Program
- 2020 Fall **Introduction to Algorithms**, teaching assistant, University of Michigan Ann Arbor

Other Academic Activities

- 2020 Spring **Graphing Monomial Relations** Supervised by Jeffrey Lagarias, Patric Boland
To determine the Hilbert function of monomial ideals generated by different seeds, deduced the quantitative recurrences of consecutive levels, graphed the complementary set of monomial ideals by degree, and explored the geometric properties of relation-reflecting graphs.
- 2020 Fall **Rational Points on the Boundary of Quantitative Constraints on the Hexagon**
Supervised by Harry Richman
Formulated the bijective map from $\mathbb{Q}^2$ to rational value configurations of a hexagon in $\mathbb{RP}^4$ and characterized the topological shape of angle constraints. We found out the bijection between mountain-valley configuration division in null cone space and corresponding rational parameters.
- 2020 Spring **Directed Reading Program** Supervised by Aleksander Horawa
David A. Cox's *Primes Of The Form $p = x^2 + ny^2$*
- 2019 Fall **Graph Coloring: Euler Color Algorithms** Supervised by Seth Pettie
Motivated by Szekeres-Wilf algorithm, studied the edge-coloring algorithms using Vizing Theorem's color fan as its algorithmic tool and randomness to reduce analyzed running time and sublinear vertex-coloring algorithms.